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Stress evolution due to electromigration in confined metal lines

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Electromigration is an important concern in very large scale integrated circuits. In narrow, confined metal interconnects used at the chip level, the electromigration flux is resisted by the evolution of mechanical stresses in the interconnects. Solutions for the differential equation governing the evolution of back stresses are presented for several representative cases, and the solutions are discussed in the light of experimental as well as theoretical developments from the literature.

I. INTRODUCTION

For most metals of current interest as interconnects in microelectronic devices, electromigration means mass flow of constituent atoms accompanying the current of electrons.[1] Electromigration (EM) is of a primary concern in narrow line metallizations on semiconductor chips where the current density can be extremely high, $>10^5$ A/cm^2, and where it can lead to severance of the lines.[1] In confined metal interconnects, deposited on an oxidized silicon substrate and covered by a dielectric passivation layer, electromigration gives rise to stresses which can enhance as well as retard electromigration.[2] The relatively good EM resistance of confined metal lines is ascribed to the retarding action due to the stress gradients caused by electromigration. The first observations of the effect,[3] suggested that there is a critical line length below which the EM flux can entirely be balanced by the stress directed counterflux of atoms.[2–4] Experimental analyses of the material transport and measurements of stresses arising due to electromigration suggest that the steady state of constant, or zero, flux can be achieved relatively fast;[3,5] hence generally only steady states have been considered in theoretical treatments.

Recently, however, Ross[6] and Kircheim[7] have modeled transient stress buildup during electromigration. Ross' model is essentially numerical while Kircheim proposes a physically based model involving two coupled differential equations. We have proposed a physically based, analytic EM model, which is capable of describing stress evolution in realistic EM conditions.[8–10] In this article we present a detailed derivation of our model starting from basic physical notations, after which we apply the model to analytic estimation of stress buildup during EM. Finally we show by numerical solutions that the analytic estimates given are generally quite sufficient to provide an illuminating, semiquantitative picture of stress evolution during electromigration of confined interconnect lines.

II. ELECTROMIGRATION AND STRESS DIRECTED FLUX OF ATOMS

We consider a thin and narrow interconnect line, deposited on an oxidized silicon substrate and covered by a rigid dielectric passivation layer, as depicted schematically in Fig. 1. For the sake of simplicity, we model the lines in question as embedded in an infinitely thick, rigid dielectric continuum. Also, for definitiveness in numerical examples, we use the material parameters of pure aluminum for the metallization and of silicon for the rigid surrounding continuum.

When an electric current passes through a polycrystalline thin film interconnect, the atomic flux at moderate temperatures comes primarily from electromigration along grain boundaries.[1] In this article we restrict ourselves to the technologically important case when the thin interconnect metallizations have columnar grain structure, so that all grain boundaries are perpendicular to the substrate, as depicted in Fig. 1. In the following we assume that material transport along the interconnect line is effected by grain boundary diffusion alone. When the grain boundary diffusion coefficient is denoted by D_{GB}, the effective diffusion coefficient, normalized to correspond to the entire cross section of the line, becomes

$$D_a=\delta D_{GB}/d, \tag{1}$$

where δ is the grain boundary width and d is the grain size.[1]

There has been a continuous trend to miniaturize the features on a chip; in the most advanced microelectronic devices of today the height and width of interconnect lines have entered submicron range. Submicron size interconnect lines tend to have "near-bamboo" grain structure with only a few boundaries along the line direction.[1] Therefore, the mass transport in them is dominated by bulk diffusion. Electromigration in the near-bamboo interconnects will be discussed elsewhere.[11]

Diffusive fluxes of atoms arise due to potential differences between different locations of the interconnect line. Here we consider only the chemical and electric potentials.

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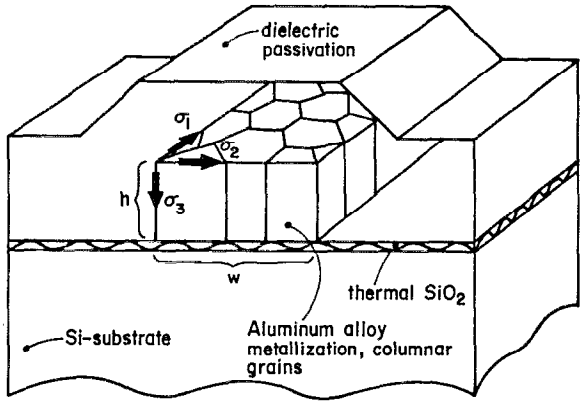


FIG. 1. Schematic of a confined interconnect line with a columnar grain structure. Also shown are the principal stresses, σ_1 along the line direction, σ_2 in the width (w) direction, and σ_3 in the thickness (h) direction.

For atomic diffusion the chemical potential function in question μ was shown by Herring[12] to be the difference between the atomic and vacancy chemical potentials

$$\mu=\mu_a-\mu_v. \tag{2}$$

In the case that the vacancies are in thermal equilibrium $\mu_v=0$ the chemical potential function is $\mu=\mu_0-\Omega\sigma$, where σ is the tensile stress acting across the grain boundary.[12]

The flux of atoms due to chemical potential gradient $\nabla\mu$ and the electric field $\mathbf{E}$ is given by

$$J_a=-D_aC_a/kT(\nabla\mu+Eq^*), \tag{3}$$

where C_a is the atomic concentration, D_a the self-diffusion coefficient, Eq. (1), and q^* the absolute value of the effective charge. Here we have taken the positive x-axis direction to point in the direction of the electron current, Fig. 2. The effective charge $q^*=|Z^*|e$, where e is the elementary charge and Z^* the effective charge number, is a poorly known quantity for aluminum at typical EM testing temperatures; a recent analysis by Tu suggests that Z^* ranges from -4 to -8.[2]

Consider now the net number of atoms $\mathrm{div}(J_a)dt$ entering a volume element (assumed to be large enough to include vacancy sources and sinks) of the interconnect line in time dt. These atoms can be spent in changing the vacancy concentration dC_v or be deposited on grain boundaries or lattice dislocations. Presently, any longer range material transport along the interconnect line is dominated by grain boundary diffusion, and we assume that atoms are deposited predominantly at grain boundaries.

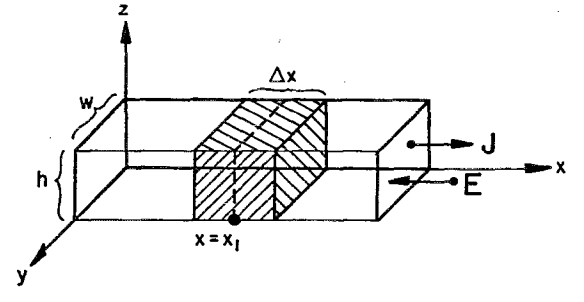


FIG. 2. Schematic of electromigration in a confined interconnect line. The atomic flux J is in the positive direction of the x axis when the electric field E points towards the origin.

Because of the confinement by the surrounding metallization, attached to the rigid dielectric, the deposition creates lateral stresses σ_1 and σ_2, Fig. 1. Also, because of Poisson contraction, σ_3 arises in the z direction. In thermal equilibrium, the chemical potential in any cross section is constant at all grain boundaries, independent of their orientation. This implies that atoms are deposited at differently oriented grain boundaries such that the lateral tensile stresses across the boundaries become equal, i.e., $\sigma=\sigma_1=\sigma_2$. The local equilibrium at grain boundaries in any particular cross section will be reached in a time span that is much shorter than required for longer range diffusion along the interconnect line. This simplifies our analysis of stress evolution during electromigration, because we need to be concerned with only one stress component driving the stress-directed flux along the interconnect line.

Let us illuminate the above point by a numerical example. The diffusion distance at a time t can be qualitatively estimated as $L\approx\sqrt{D_at}$, see e.g., Ref. 13. A typical diffusion distance in a cross section is on the order of the grain size, say $d\approx0.5$ μm at most. On the other hand, we are interested in stress evolution along the interconnect line, typically on a scale that lies on the order of 5 μm or more. Hence we can readily estimate that the local stress equilibrium over a cross section at x will be reached in one hundredth of the time that is needed for atoms to diffuse from x to $x+5$ μm along the line.

We assume a uniform stress $\sigma=\sigma(x)$ over any cross section, defined by the position coordinate x along the line, Fig. 2. Let the climbing grain boundary (GB) dislocations change the number of available lattice sites per unit volume by dC. In confined metal lines the relative density change dC/C corresponds to the stress increment as

$$\frac{dC}{C}=\frac{-d\sigma}{B}, \tag{4}$$

where B is the applicable modulus. We show in Appendix A that for aluminum lines confined in a silicon matrix B ranges from 0.5 to 0.75 times the Young's modulus for aluminum, depending on the line geometry. The concentration $C=C(x)$, should be understood to be a volume average over $hw\Delta x$, Fig. 2, as discussed in Appendix A.

The continuity equation in the line direction gives now the material balance as

$$\frac{-\partial J_a}{\partial x}=\frac{\partial C_v}{\partial t}+\left(\frac{C}{B}\right)\frac{\partial\sigma}{\partial t}, \tag{5}$$

where we have substituted dC from Eq. (4); here C_v should be understood as a volume average over $hw\Delta x$, Fig. 2, similarly to C, see Appendix A. We assume with Blech and Herring[4] that the vacancies are in equilibrium with the stress, even inside the grains. The alternative proposition

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that sinks/sources are not able to maintain the vacancy equilibrium is discussed in Appendix B. At equilibrium $\mu_v \approx 0$ or

$$C_v = C_{v0} \exp(W_f + \Omega\sigma/kT), \tag{6}$$

where Ω is the atomic volume, kT the thermal energy, W_f the interaction energy between the vacancy and the stress field, while C_{v0} is the vacancy concentration in the absence of any stress effects.[14] The interaction energy W_f depends on the hydrostatic stress p as $W_f = pV_f$, where V_f is the relaxation volume of the vacancy. For the sake of simplicity, we take $V_f \approx 0$ because this will not affect our results. Substituting C_v to Eq. (5) we readily find

$$\frac{-\partial J_a}{\partial x} = \left(\frac{C}{B} + \frac{C_v \Omega}{kT} \right) \frac{\partial \sigma}{\partial t}. \tag{7}$$

The relative number of atoms involved in restoring the vacancy equilibrium and in creating the stress is given by the ratio of the two right-hand side (RHS) terms $(C_v/C)(B\Omega/kT)$. At a typical EM testing temperature, 200 °C, the term $B\Omega/kT$ is of the order of 120 for aluminum when we take B to be on the order of 50 GPa. The vacancy concentration C_v/C is typically on the order of 10^{-7} at 200 °C, and much lower at room temperature. Accordingly, the second term on the right hand side of Eq. (7) is negligible. Consequently at the sites of flux divergencies, climbing GB dislocations consume practically all the net flux in creating mechanical stresses, while GB as well as lattice dislocations need to move only infinitesimal distances to resume the vacancy equilibrium.

In the case of vacancy equilibrium, we get from Eqs. (3) and (7) for stress evolution along the interconnect line during electromigration

$$\frac{\partial \sigma}{\partial t} = \frac{\partial}{\partial x} \left[\kappa \left(\frac{\partial \sigma}{\partial x} + G \right) \right], \tag{8}$$

under the fair approximation that $C_v/C \ll B\Omega/kT$ and $C = C_a = 1/\Omega$. Here we have denoted $\kappa = D_a B\Omega/kT$ for the diffusivity and introduced the shorthand notation $G = Eq^*/\Omega$.

Equation (8) is formally very similar to the familiar partial differential equation for diffusion (e.g., Ref. 13), with the diffusivity κ replacing the conventional diffusion coefficient D_a. We can readily find solutions for several electromigration problems based on this analogy. We shall first suppose that the diffusion coefficient D_a is effectively constant and present analytic solutions for Eq. (8) in several cases. Later we relinquish this condition and show that the analytic results obtained agree reasonably well with the numerical solutions with variable D_a.

Equation (8) is also very similar to Eq. (B1) of Appendix B, the so-called "electromigration equation," dealt with, e.g., by Rosenberg and Ohring,[15] Shatzkes and Lloyd,[16] Nakajima *et al.*,[17] and most recently by Clement and Lloyd.[18] The formal similarity becomes more apparent if the stress in Eq. (8) is expressed in terms of C_v through Eq. (6). Physically, however, Eqs. (8) and (B1) are widely different. According to Eq. (8), the current-induced flux creates a stress-directed counterflux, which retards EM damage, while according to Eq. (B1) the counterflux arises due to the vacancy concentration gradient. The remarkable point is that significant vacancy concentration gradients can be created in a matter of seconds because of the almost negligibly small material transport involved, as detailed in Appendix B. On the other hand, Eq. (8) predicts that stress evolution in interconnect lines during electromigration can last hundreds of hours as to be dealt with below.

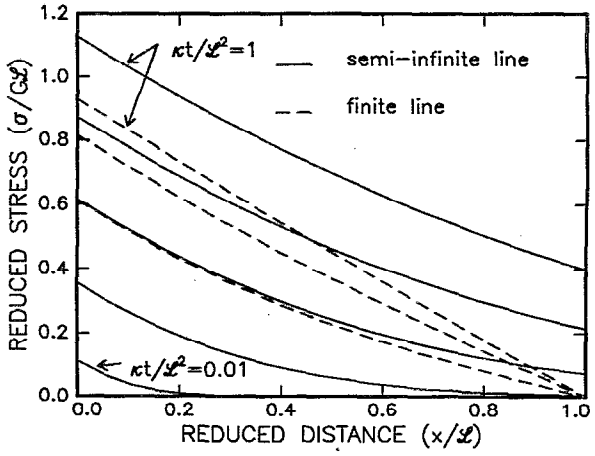


FIG. 3. Tensile stress evolution in semi-infinite (solid line) and finite lines (dashed line), constant D_a.

III. SOLUTION FOR A SEMI-INFINITE LINE (D_a CONSTANT)

Consider a semi-infinite line, where the flux is blocked at the end $x=0$. Physically these blocking boundaries might be due to bamboo grains, with no boundaries along the line direction,[19] or due to tungsten studs used in the semiconductor industry.[1] The solution of Eq. (8) for the semi-infinite case, under the boundary condition that there is no flux through the blocking boundary, $J_a = 0$ for $x = 0$ (i.e., no voids form here), can be found by using the technique of Laplace transformations[20]

$$\sigma = G[\sqrt{(4\kappa t/\pi)} \exp(-x^2/4\kappa t) - x \operatorname{erfc}(x/\sqrt{4\kappa t})]. \tag{9}$$

It should be noted that $G = Eq^*/\Omega$ gives the stress gradient at $x=0$, and provides a natural unit to measure the stress. In presenting the data we are going to make use of the reduced stress, distance, and time defined as $\sigma/G\mathscr{L}$, $x/\mathscr{L}$, and $\kappa t/\mathscr{L}^2$, where $\mathscr{L}$ is an appropriate characteristic length, chosen to reflect the physical situation. Some stress distributions at different times are shown in Fig. 3 (solid lines). We have chosen $\mathscr{L}$ to be the physical diffusion distance of the corresponding finite line, to be dealt with later. Presently, $\mathscr{L}$ can be interpreted simply as the length over which we consider the stress buildup in the semi-infinite line. Particularly for $x=0$, Eq. (9) reduces to the simple form

$$\sigma = 2G\sqrt{(\kappa t/\pi)}, \tag{10}$$

where G is positive for the electron current directed outward from the boundary and negative for the electron cur-

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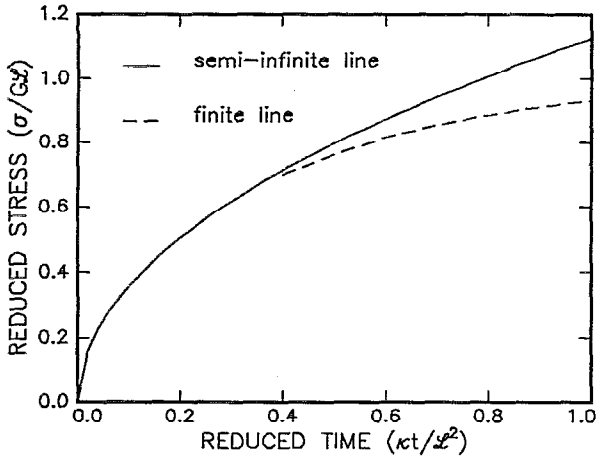


FIG. 4. Stresses at the blocking boundary in semi-infinite (solid line) and finite lines (dashed line) as a function of time, constant D_a.

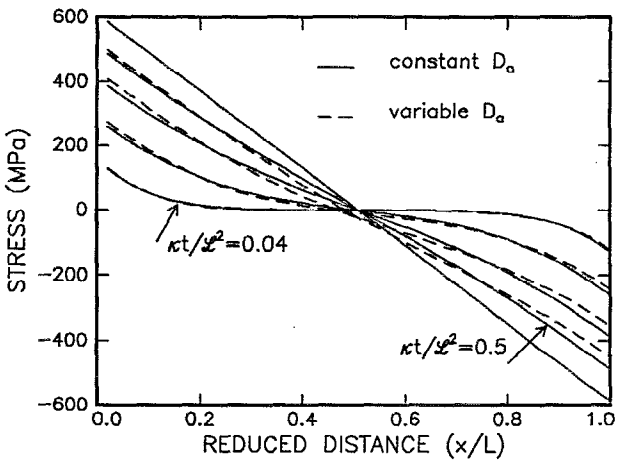


FIG. 5. The stress evolution in the finite line at 200 °C, in the case of constant (solid line) and variable (dashed line) diffusion coefficient D_a. A real stress scale is used, because variable D_a depends on the absolute stress level.

rent toward the boundary. The stress increases in the square root of time, as shown in Fig. 4 (solid line). By using reasonable values for aluminum $B\approx 5\times 10^4$ MPa, $G\approx 16.5$ MPa/μm and $D_a\approx 3\times 10^{-16}$ m^2/s (see Sec. VII), we find that a stress level of about 200 MPa is achieved in an hour at 200 °C. In case there is leakage flux through the boundary at $x=0$, Eqs. (9) and (10) still apply; we have only to redefine the stress gradient at $x=0$ as $G=(1-f)Eq^*/\Omega$, where f is the fraction of the EM flux that penetrates the boundary. This is a natural consequence of the fact that the stresses are proportional rather to the flux divergence than the absolute value of the flux.

According to Eq. (9) there is no natural upper limit for the stress; i.e., there is no steady state. However, it is apparent that the stresses will eventually reach the yield strength, which leads to the establishment of a "forced" steady state. We suppose that ideal plasticity occurs at the yield stress σ_y. From material balance it follows that at the steady state the boundary between the elastic and plastic regions recedes from the origin at a constant velocity of $v=G\kappa/\sigma_y$. The solution of Eq. (8) in the moving coordinate system $x'=x-vt$, under the condition that $\sigma=\sigma_y$ at $x'=0$, is easily found to be

$$\sigma=\sigma_y\exp(-vx'/\kappa)=\sigma_y\exp(-Gx'/\sigma_y), \tag{11}$$

for $x'>0$. It should be noted that the total flux is zero in the plastic region $x'<0$; i.e., the EM flux is entirely balanced by the counterflux due to plastic flow. In the transition region $\sigma_y>\sigma>0$ the total flux gradually increases from zero to the maximum value due to electromigration alone.

IV. SOLUTION FOR A FINITE LINE (D_a CONSTANT)

Consider a line of length L, where the flux is blocked at both ends. The steady state solution is readily found by substituting $\mu=\mu_0-\sigma\Omega^4$ and setting $J_a=0$ in Eq. (2). The result, originally given by Blech and Herring,[4] is

$$\sigma=GL[1/2-(x/L)]. \tag{12}$$

The maximum stress at $x=0$ and L is $|\sigma|=GL/2$, which by using reasonable values for aluminum of $G\approx 16.5$ MPa/μm, corresponding to a current density of about 1 MA/cm^2 at 200 °C, and $L=50$ μm, becomes about 400 MPa. This value is quite feasible considering the actually measured stress levels in confined Al interconnect lines after cooldown from heat treatment at elevated temperatures, on the order of 350–600 MPa.[21–23]

Consider then the time-dependent solution for a finite line, when there is no flux through the ends at $x=0$ and L. By the technique of Laplace transformations it is found[20]

$$\sigma=GL\left(1/2-\xi-4\sum_{n=0\to\infty} m_n^{-2}\exp(-m_n^2\tau)\cos(m_n\xi)\right), \tag{13}$$

where we have used the reduced length $\xi=x/L$ and time $\tau=\kappa t/L^2$ and introduced the shorthand notation $m_n=(2n+1)\pi$.

Figure 5 (solid lines) shows the stress evolution according to Eq. (13): the antisymmetric stress distributions arise at the opposite blocking boundaries, move inward, and eventually merge in the middle of the line. Obviously, the physical diffusion distance is one-half of the line length $\mathscr{L}=L/2$. The steady state for the finite line is practically achieved in a time of $t\approx\mathscr{L}^2/\kappa$. This may be compared with the common definition of the diffusion distance, $\mathscr{L}\approx\sqrt{D_a t}$, see e.g., Ref. 13, remembering that presently κ plays the same role as D_a usually does. Figure 3 compares the solutions for the finite and semi-infinite line: at small times the solutions agree closely. Figure 4 compares the stress evolution at the blocking boundary for the finite and semi-infinite line: up to the time $t\approx\mathscr{L}^2/\kappa$ the behaviors differ so little that the semi-infinite solution can be used to a fair approximation when estimating the stress levels.

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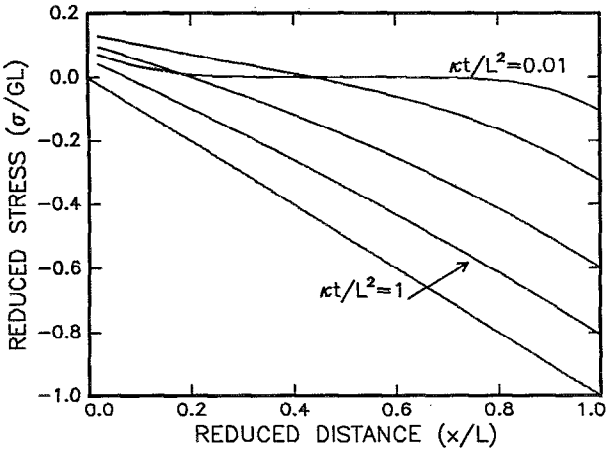


FIG. 6. Stress evolution in the presence of a void ($\alpha=5$) in a finite line, constant D_a.

V. SOLUTION FOR A FINITE LINE IN THE PRESENCE OF A VOID (D_a CONSTANT)

In many cases of practical interest, the interconnect lines already contain numerous, thermal stress-induced voids after manufacturing.[22,24] According to a simple analysis,[24] we can write for the flux from the void to a connecting grain boundary

$$J_{\text{void}}=(D_a/kT)\sigma/w, \tag{14}$$

where w is the linewidth and σ the stress normal to the boundary. We have estimated that the extent of the stress gradient is on the order of the grain size d, and normalized the flux to correspond to the entire cross section to facilitate the comparison to the EM flux. Consider the case when the void at the line end is the only source of the flux: Eq. (14) becomes then analogous to the radiation boundary condition of heat conduction.[17] We now assume that there is a void in the line at $x=0$, the flux from which is given by Eq. (14). At the other end of the line there is a blocking boundary $J_a=0$ at $x=L$. Under these conditions the solution of Eq. (8) can be found analogously to the techniques described by Carslaw and Jaeger for the case of heat conduction,[20]

$$\sigma=-G\xi+2GL \times \sum_{n=0\to\infty} \frac{[\beta_n\cos(\beta_n\xi)+\alpha\sin(\beta_n\xi)]\exp(-\beta_n^2\tau)}{(\alpha+\alpha^2+\beta_n^2)\beta_n} \times\{\cos(\beta_n)+\beta_n\sin(\beta_n)-1 +\alpha[\sin(\beta_n)-\beta_n\cos(\beta_n)]/\beta_n\}. \tag{15}$$

Here we have used the same reduced units, $\xi=x/L$ and $\tau=\kappa t/L^2$, as in Eq. (13) and denoted $\alpha=L/w$. The terms β_n are the roots of the transcendental equation

$$\beta\tan\beta=\alpha, \tag{16}$$

tabulated, e.g., but Carslaw and Jaeger.[20] Figure 6 gives some solutions of Eq. (15) for $\alpha=5$, which roughly corresponds to the experimentally observed void separation in pure aluminum interconnects after sintering treatment.[24] Next to the void the maximum stress $GL/\alpha=Gw$ is reached during the transient; most usually it is only a fraction of the compressive steady state stress at the blocking boundary GL. In the steady state, the stress is compressive everywhere except next to the void where it is essentially zero.

VI. SOLUTIONS IN THE CASE OF STRESS DEPENDENT DIFFUSION COEFFICIENT D_a

The self-diffusion coefficient D_a for atoms depends on the vacancy concentration according to Eq. (6). Additionally, the vacancy diffusion coefficient D_v depends on the hydrostatic pressure p as

$$D_v=D_v(0)\exp(-pV_m/kT), \tag{17}$$

where V_m is the dilatation associated with vacancy migration and $D_v(0)$ is the vacancy diffusion coefficient at $p=0$.[14] From the above and Eq. (6) we find for the self-diffusion coefficient

$$D_a=D_a(0)\exp(\Omega\sigma/kT)\exp[-p(V_f+V_m)/kT], \tag{18}$$

where $D_a(0)$ is the self-diffusion coefficient in the absence of any stress effects. Because for vacancies V_f is small and negative (contraction of the lattice site) while V_m is small and positive, the pressure dependence partly cancels out. Measurements of the apparent activation volume for self-diffusion in aluminum give results on the order of Ω for both lattice and grain boundary diffusion,[25] which shows that V_f+V_m is small. In the following we take $V_f+V_m\approx 0$ as a reasonable first order approximation.

The numerical integration of Eq. (8) with a stress dependent D_a, Eq. (18), was accomplished by the method of finite differences; generally 100–200 divisions in the length scale produced quite accurate results when checked against known analytic solutions. The time step was chosen small enough to avoid oscillations in integration; alternatively Crank–Nicholson algorithm[26] was used for long times. The numerical results turned out to be not too different from the analytic solutions with a constant D_a.

This may not appear too surprising in the case of finite lines when it is realized that the steady state $J_a=0$ is independent of the diffusion coefficient, see Eq. (2). So the numerical solutions are constrained to evolve from the same initial state to the same steady state as the corresponding analytic solutions in the case of constant D_a. Some numerical solutions for the finite line with blocking boundaries at $x=0$ and L are shown in Fig. 5 (dashed line). In the calculation the temperature was selected to be 200 °C, and the factor $GL/2$ was chosen to produce the maximum stress $|\sigma|=600$ MPa at the blocking boundaries. The evolution of stresses is practically symmetric up to stress levels of about 300 MPa, after which the asymmetry due to the stress dependent D_a begins to show. Nevertheless, it appears that analytic solutions of Eq. (8) with a constant diffusion coefficient provide a useful basis to estimate stress evolution during electromigration.

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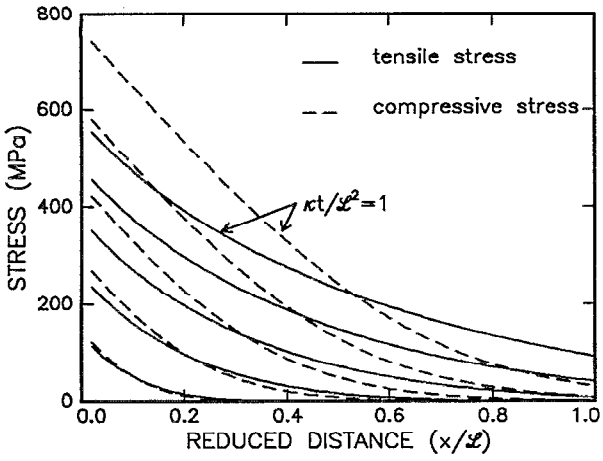


FIG. 7. Stress evolution in a semi-infinite line at 200 °C in the case of variable diffusion coefficient D_a, for compression (solid line) and tension (dashed line).

Perhaps somewhat more surprisingly, numerical results with a stress dependent D_a for the semi-infinite line as well turn out to be reasonably close to the analytic estimates with a constant D_a, Eqs. (9) and (10). Figure 7 shows the tensile stress buildup (dashed line) as well as the compressive stress buildup when the current direction is changed, for the semi-infinite line at 200 °C. The stress for a constant D_a, Fig. 3 (solid lines), would lie halfway between the cases shown in Fig. 7. At the blocking boundary, the absolute values of the numerically calculated stresses differ by 15%, at most, from the analytic estimates with a constant D_a. The stress evolution at the blocking boundary is still described fairly well by Eq. (10). A closer analysis shows that instead of the time exponent 0.5, Eq. (10), an exponent of 0.45 for compression and 0.55 for tension would describe the stress data of Fig. 7 marginally better. The physical reason for the remarkably good agreement with Eq. (10) is that stress buildup at the blocking boundary requires long-range diffusion from the regions of the line where the stress is still close to zero. Given this constraint, the stress gradient close to the boundary changes in a way that partially balances the effect of the stress dependent diffusion coefficient, see Fig. 7. For example, the tensile stress enhances diffusion and consequent stress buildup, the more the closer to the boundary: the result is a steeper stress gradient driving the backflow.

In conclusion, the analytic solutions may in several cases be well sufficient for practical estimation of stress buildup during electromigration. Moreover, relatively simple analytic solutions to Eq. (8) with constant D_a, and under different boundary conditions, can readily be found from the literature (e.g., Ref. 20). We emphasize the inherent benefit of the analytic formulation which readily displays the effects of different physical parameters, in contrast with what may be the case with numerical solutions. However, for modeling of more complicated line structures we must resort to numerical integration of Eq. (8). Eventually we should be able to model realistic metallizations, where diffusion coefficients change from grain to grain, and where there are several voids and nonuniformly distributed residual stresses inherited from manufacturing. A report of modeling of EM damage in these kind, realisticlike interconnect line structures is in preparation.[11]

VII. DISCUSSION

A. Comparison to other works

Kinsbron, Blech, and Komem have presented an essentially similar expression to our Eq. (8), although formulated rather in terms of concentration than stress.[5] They supposed that the stresses at the blocking boundaries take immediately the maximum value, after which the stress gradients only work their way inwards the line. The treatment results in the right steady state, which they were mainly interested in. However, the present solution for the finite line, Figs. 3 and 5, shows that the maximum stresses buildup in concert with the inward propagation of the stress front.

Shatzkes and Lloyd use Eq. (B1) of Appendix B to model vacancy supersaturation with time at the blocking boundary.[16] By supposing that nucleation takes place after reaching a critical vacancy supersaturation, they demonstrate that EM lifetime depends in -2 power on the current density. In the light of present results it appears more likely that when the EM lifetime is nucleation limited, nucleation results from stress evolution at the sites of flux divergencies. The same current density exponent, -2, is predicted for the latter case, too, as to be discussed below.

Kircheim presents two coupled differential equations, a continuity equation for vacancy concentration, and a rate equation for sink and source reactions.[7] In the limit, when sink and source reactions are fast, these equations reduce to an equation, which is essentially similar to our Eq. (8). Kircheim considers a finite line blocked at both ends, and finds that his coupled equations predict a quasi-steady state at short times, where the vacancy concentration remains approximately constant with time. Based on the line of argument detailed in Appendix B the reason for the occurrence of the quasi-steady state appears to be the following: When EM starts from the vacancy equilibrium at zero stress, there is no driving force for the sink or source reactions. Consequently, during a short period of time the vacancy distribution follows Eq. (B1), tending towards the steady state given by Eq. (B2). On the other hand, if the sink and source processes were infinitely fast, the stress evolution should follow Eq. (8) at all times, and the vacancy concentration would respond to the stress according to Eq. (6). The quasi-steady state appears then simply as a transition between these two behaviors, the first one being due to initial, fast vacancy redistribution and the second one due to eventual, relatively slow stress buildup.

Recently Hemmert and Costa proposed to use beta degradation of transistors on the chip as the stress gauges; with these they were able to follow stress evolution at the compression end of long interconnect lines, up to 2 mm.[27] For practical purposes these lines certainly qualify as semi-

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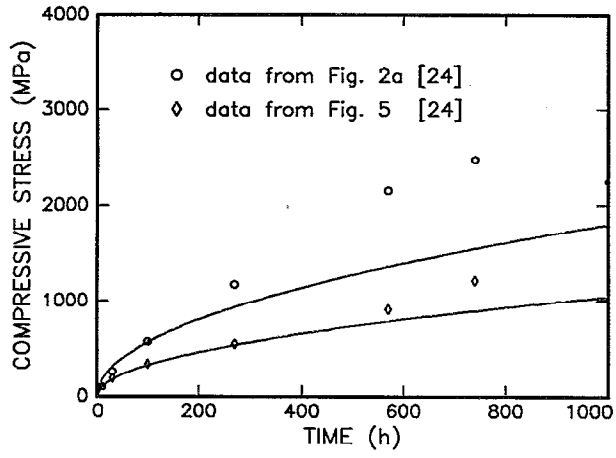


FIG. 8. The stress evolution results of Hemmert and Costa (Ref. 27) at the blocking boundary, with the fit to Eq. (10).

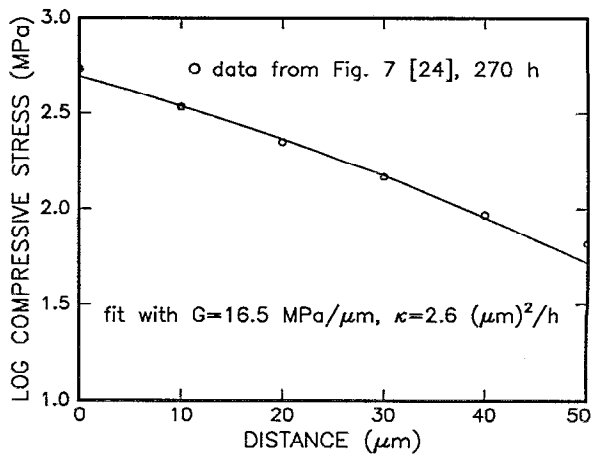


FIG. 9. The stress distribution measured by Hemmert and Costa (Ref. 27) after $t=270$ h current loading, with the fit to Eq. (9).

infinite, and we can apply our Eqs. (9)–(11) to interpret their results.

Hemmert and Costa followed the stress evolution at 200 °C under 1 MA/cm^2 EM current density at the compression end of several lines. We have replotted their data (as converted to an absolute stress scale) in Fig. 8 for two samples.[27] Up to the stress level of about -700 MPa, reached in 100–500 h, their results agree remarkably well with our square root law, Eq. (10). However, after this limit, the data points deviate successively more from the square root law. It is remarkable that the deviation takes place for both samples at about the same stress level, suggestive of reaching a common critical stress. Hemmert and Costa presume that fracturing of the passivation layer renders the nearby stress gauge inoperative, resulting in anomalously high beta-degradation readings.[27] From the fit of Eq. (10) to the data, Fig. 8, we arrive at the estimates for the quantity $G\sqrt{\kappa}$, 50 MPa/$\sqrt{\text{h}}$, and 30 MPA/$\sqrt{\text{h}}$, for the upper and lower curves, respectively. If we could estimate κ independently, we would be able to calculate the effective charge number Z^* based on the known value of $G\sqrt{\kappa}$.

Hemmert and Costa demonstrated also that the logarithmic stress decreases approximately linearly with the distance from the blocked line end. We have reproduced the stress distribution they found after 270 h under current loading in Fig. 9. We deemed that this data should most accurately reflect the prediction of Eq. (9): the stress level is high enough to yield sufficient stress resolution, while still below the level for cracking the passivation. The fit according to Eq. (9) produces the parameters $G=16.5$ MPa/μm and $\kappa=2.6$ $(\mu\text{m})^2$/h.

Because the effective diffusivity is $\kappa=D_aB\Omega/kT$, we can calculate D_a from the value of κ obtained. By estimating as before $B\approx 50$ GPa, we find for the effective grain boundary diffusion coefficient $D_a\approx 3\times 10^{-16}$ m^2/s. By using a grain size of $d\approx 1$ μm, we estimate a grain boundary diffusion coefficient of $\delta D_{GB}\approx 3\times 10^{-22}$ m^3/s, which compares well with the value $\approx 1\times 10^{-23}$ m^3/s given by Kaur *et al.* for aluminum at 200 °C.[25] Most remarkably, based on the estimated value of $G=Eq^*/\Omega$, we can calculate the corresponding effective charge q^*. The result is $Z^*=-3.5$, which is close to the lower bound (-4) estimated by Tu.[2] The data of the lower curve of Figs. 8 and 9 correspond to the same sample. The values G and κ shown in Fig. 8 were determined from the distribution of the stress with the distance, while $G\sqrt{\kappa}$ for Fig. 9 was obtained from the change of the stress with time: these two data sets are independent. It is remarkable that the values of $G\sqrt{\kappa}$ determined from both data sets agree within about 10%, which shows the consistency of our estimates. In conclusion, the data by Hemmert and Costa strongly suggest that the effective charge number for Al (4% Cu) is around -4 at 200 °C, i.e., significantly smaller than is commonly held.[1,2]

B. Void nucleation and yielding

In a finite line it is possible to achieve the steady state before yielding. The total flux becomes zero when the stress at the end reaches the value $GL/2$, Eq. (12); at this point the EM flux will be entirely balanced by the counterflux, and consequently there will be no further EM damage. This distinct transition in the accumulation of EM damage was first observed by Blech.[3] He found a critical current density, as well as a critical line length, below which electromigration eventually stopped.

If $\sigma_y < GL/2$, the blocked boundary regions yield before the stress gradient reaches G in the central portions of the line. At the eventual steady state the stress gradient becomes G also in the central line up to the location where the stress reaches the yield strength σ_y; after this the stress remains at σ_y until the blocking boundary. In the picture of ideal plasticity, when $\sigma=\sigma_y$ the plastic flow balances the EM flux.

Grain boundary sliding can nucleate voids in the yielded interconnect lines, in analogy with the thermal stress-induced void nucleation during cooldown from a passivation or sintering treatment.[28] On the compression side the high stresses and grain boundary sliding can eventually lead to fracturing of the passivation layer, and to

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hillock growth. *In situ* electron microscopic studies of EM show evidence of extensive grain boundary sliding before the first voids become visible.[29]

If voids begin to nucleate, or the passivation cracks at σ_y we see from Eq. (10) that the time to reach this stage is

$$t_y=(\pi/\kappa)(\sigma_y/2G)^2. \tag{19}$$

Hence the time to yielding or cracking depends in the power of -2 on the current density. In the case of a finite line, the absolute value of the exponent is always equal or larger than 2, depending on the stress level (or line length), see Fig. 4. In the case the critical stress level lies close to the maximum steady state stress, the current exponent approaches infinity.

C. Stress evolution in the presence of voids

The steady state value of the stress next to a void is close to zero at the steady state. If there are several voids in the line with only relatively short distances between them, the stress keeps initially small over the entire length of the line. However, eventually the voids on the compression side will be filled up with atoms, making the distance from the blocking boundary to the nearest void relatively large at the compression end. At the steady state then, the stresses are close to zero at voids and about $-GL$ at blocking boundaries; the atomic flux goes from the voids to compressively stressed blocking boundaries. Therefore, in voided metallizations all significant stresses will eventually be compressive as also experimentally observed.[27] In the case of an entirely blocking boundary, the stress gradient reaches eventually the level G needed to stop the EM flux, Eq. (12). If yielding occurs before this stage, a regime of constant stress $\sigma=-\sigma_y$ forms next to the blocking boundary, as discussed above.

VIII. CONCLUSIONS

(1) EM flux builds up stresses in the case there is a vacancy equilibrium with the stress. It appears that no large vacancy supersaturations can arise in the presence of dislocations and internal boundaries.

(2) In long enough lines stresses at blocking boundaries can reach the yield point, where voids can be nucleated by plastic deformation.

(3) The stress at the blocking boundary increases approximately as the square root of time. The time to yield depends in the power of -2 to $-$infinity on the current density, depending on the line length.

(4) The effective charge number Z^* appears to be on the order of -4 during long term current loading of aluminum at 200 °C.

(5) The stresses next to voids are generally small. Because any existing voids on the compression side will eventually be filled up, in void containing lines the macroscopically observable stresses are compressive.

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APPENDIX A

The stresses created by deposition of atoms at an isolated grain boundary lying in a homogeneous matrix, have been dealt with by Weertman,[30] Raj,[31] and Vitek.[32] However, presently we are concerned with a more complicated case where

(i) the atoms are deposited simultaneously at several, differently oriented grain boundaries of the metallization, and

(ii) the ensuing stresses are shared by the metallization and the surrounding dielectric.

The extension of the previous treatments to cover the present case would involve a complicated set of differential equations, for which only numerical solutions would be possible.

An analytic model which can address the above questions, even if an oversimplification, is provided by the Eshelby theory of inclusions.[33] The Eshelby theory seeks to find the stresses arising due to uniform free strains, as due to phase transformations, for example, in an inclusion residing in an infinite matrix. Presently, the free strains are the dilational strains ϵ_1^T, ϵ_2^T, ϵ_3^T due to deposition of atoms at grain boundaries, and the inclusion is the interconnect line. In particular, in the case where the interconnect line can be modeled as an elliptic cylinder, the uniform free strains give rise to uniform stresses in the interconnect.[33] Of course, the interconnect lines are rectangular rather than elliptic in cross section. However, for the same dilational free strains, solutions for rectangular and elliptic lines show that the resulting dilational stresses agree closely.[34]

Hence, we shall model the interconnects as elliptic cylinders of aluminum, lying in an infinite silicon matrix. We denote the number of atoms per unit volume by C. Let now dC atoms be deposited at grain boundaries so that a uniform lateral stress σ develops, as discussed in Sec. II. We define the change in the volume free strain due to deposition as

$$d\epsilon^T=d\epsilon_1^T+d\epsilon_2^T+d\epsilon_3^T=dC/C. \tag{A1}$$

As usual in continuum models, for Eq. (A1) to be meaningful, we must consider C to be an average over a sufficiently large volume, so as to produce a uniform $d\epsilon^T$. We assume small enough grain size such that the concentration, when averaged over the line volume $hW\Delta x$ around the position x, Fig. 2, results in well-defined values, $C=C(x)$ and $dC=dC(x)$. The same considerations apply also to the vacancy concentration $C_v(x)$, which must be understood as a similar volume average.

The free strains and dilatational stresses can be shown to be connected as

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TABLE I. The coefficients $b=\sigma_3/\sigma$ and B/E^*, for an aluminum interconnect line embedded in Si and SiO_2 matrix, for various aspect ratios h/w.

	h/w	0	0.25	0.5	0.75	1
Si	b	0	0.19	0.32	0.42	0.48
	B/E^*	0.75	0.71	0.65	0.59	0.53
SiO_2	b	0	0.17	0.30	0.40	0.49
	B/E^*	0.75	0.63	0.53	0.45	0.39

$$\epsilon_i^T = \sum_{j=1}^{3} T_{ij}^* \sigma_j, \quad (i=1,2,3), \tag{A2}$$

where $T^*=T+s-s^*$.[21] Here T is a transformation matrix given below, while s and s^* denote the elastic compliances of silicon and aluminum, respectively. The transformation matrix is

$$T=\frac{1}{E}\begin{bmatrix} -1 & \nu & \nu \\ \nu & -1-2(1-\nu^2)h/w & 1-2\nu^2 \\ \nu & 1-2\nu^2 & -1-2(1-\nu^2)w/h \end{bmatrix}, \tag{A3}$$

where $E=1/s_{11}$ and $\nu=-s_{12}/s_{11}$ are the Young's modulus and Poisson's ratio of silicon, and h/w is the aspect ratio.[24] We are interested in the volume free strain, Eq. (A1), for which we obtain from Eqs. (A1) and (A2)

$$\epsilon^T = \sum_{i,j=1}^{3}\sum T_{ij}^* \sigma_j. \tag{A4}$$

We set $\sigma=\sigma_1=\sigma_2$ and $\epsilon_3^T=0$ (no grain boundaries perpendicular to the z direction) in Eq. (A2), to obtain $\sigma_3=b\sigma$, where the coefficient b depends on the elastic properties of aluminum and silicon, and on the aspect ratio h/w. Substitution of this relation in Eq. (A4) gives finally

$$d\epsilon^T = \frac{dC}{C} = \frac{-d\sigma}{B}, \tag{A5}$$

where B is a coefficient which depends only on the elastic properties of aluminum and silicon, and on the aspect ratio h/w.

Table I shows the values $b=\sigma_3/\sigma$ and B/E^*, where E^* is the Young's modulus of aluminum, for various representative aspect ratios h/w. For comparison, Table I also shows the case where the aluminum lines are embedded in a SiO_2 matrix. For the low aspect ratio of $h/w\approx 0$, B depends only on the elastic properties of aluminum, $B=E^*/[2(1-\nu^*)]$, where ν^* is the Poisson's ratio. It may also be noted that for $h/w=1$, B is about 25% larger in the case of Si matrix. Hence, for example, Eq. (8) predicts that the same current induced flux results in stronger stress-directed counterflux in the case of SiN_x passivation (the elastic properties of which are close to silicon) than in the case of SiO_2 passivation. Further, this difference should increase with an increasing aspect ratio h/w.

Strictly, the Eshelby theory of inclusions deals with free strains that remain constant over the volume of the inclusion. In the interconnects, however, the volume free strain ϵ^T becomes a function of the position x along the line during electromigration. An extension of the Eshelby theory of inclusions is capable of addressing the case when the free strains change as a power function of the position in the inclusion.[35] We do not, however, pursue this rigorous line of analysis here, but present a qualitative argument, instead, for the approximate validity of Eq. (A5) for the case of smoothly varying ϵ^T in confined interconnect lines.

It is intuitively clear that in the confined interconnect lines, the stress in any cross section, say $x=x_1$, Fig. 2, is determined mainly by the free strains in the close proximity, $x_1\pm\Delta x/2$. We estimate that Δx is on the order of the larger one of h and w.

In the linear elasticity, the free strains $\epsilon_i^T(x)$ varying linearly over the range Δx, Fig. 2, result in the same stress at $x=x_1$ as would the constant free strains $\epsilon_i^T(x_1)$. Consider now the volume free strain $\epsilon^T(x)$ and its tangent function $\epsilon^T(x_1)+\epsilon^{T'}(x_1)(x-x_1)$ at the point $x=x_1$. Equation (A5) gives a piecewisely valid estimate for any x where we can replace the function $\epsilon^T(x)$ by its tangent over a length scale which is on the order, or larger than h or w. Generally, the volume free strain varies smoothly enough for this to be at least approximately true.

APPENDIX B

Consider the case when there is no stress due to EM: i.e., the RHS stress term of Eq. (5) is negligible. For this case the chemical potential function is simply $\mu=\mu_0 - kT\ln C_v$,[14] whence Eq. (8) reduces to the form discussed by deGroot in connection to thermomigration[36]

$$\partial C_v/\partial t = D_v(\partial^2 C_v/\partial x^2 + S\partial C_v/\partial x), \tag{B1}$$

where we have introduced the shorthand notation $S=Eq^*/kT$. deGroot also presents an analytic solution for Eq. (B1) in the case of two blocking boundaries.

Equation (8) can be represented in a form similar to Eq. (B1) if we express the stress variable σ in terms of C_v through Eq. (6); we need only to replace the diffusivity D_v by $\kappa=D_aB\Omega/kT$. There is, however, a flaw in this formal similarity: while D_v is independent of C_v, D_a depends explicitly on the vacancy concentration, Eqs. (6) and (18). Physically there is an order of magnitude difference: the ratio of the respective diffusivities $\kappa/D_v \ll 1$ shows that stress evolution is a very slow process as compared to the buildup of vacancy concentration gradients in the absence of stress.

Equation (B1) is commonly referred as the electromigration equation. Nakajima *et al.* discuss this equation in connection to EM of hydrogen in vanadium;[17] Shatzkes and Lloyd consider the evolution of the vacancy concentration at the end of a semi-infinite line,[16] while Clement and Lloyd analyze the numerical solutions of Eq. (B1) under various boundary conditions.[18] Rosenberg and Ohring added a sink/source term on the RHS, and solved the steady state vacancy concentration in the vicinity of a grain boundary.[15]

The steady state solution $J_a=0$ of Eq. (B1) for a finite line blocked at both ends, is easily found to be

$$C_v/C_{v0} = SL\exp(-Sx)/[1-\exp(-SL)], \tag{B2}$$

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where C_{v0} is the equilibrium vacancy concentration and L is the line length. In the steady state the EM flux is balanced by the flux due to concentration gradient; a practical steady state is achieved in a time on the order of $L^2/2D_v$.[17] Representative numerical values at 200 °C are $D_a \approx 3 \times 10^{-16}$ m^2/s, $C_v/C_a \approx 1 \times 10^{-7}$ and $q^* \approx 4e$ (as detailed in Sec. VII). Assuming a current density of 1 MA/cm^2 we arrive at an estimate $S \approx 0.04/\mu$m. By using these values for a 25-μm-long line, we find that C_v/C_{v0} varies from 0.6 to 1.6 between the blocking boundaries; the steady state will be achieved in a time on the order of seconds. This agrees well with the conclusions of Rosenberg and Ohring.[15]

The chemical force[37] on dislocations due to above nonequilibrium vacancy concentrations, 0.6 and 1.6, corresponds to the mechanical stresses of ± 200 MPa. It is most likely that dislocations start climbing long before the chemical force reaches this high level; for comparison, aluminum is known to creep at stresses as low as 10 MPa at the temperatures in question. Because of the very small number of vacancies in relation to atoms, dislocations need to climb only infinitesimal distances in order to resume the vacancy equilibrium.

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